

PHASE 5

PROJECT DEVELOPMENT PHASE

Development Environment Setup

The environment requires a Python installation, configuration of environment variables via the `.env` file, and installation of packages defined in `requirements.txt` and `requirements-local-diffusion.txt`.

Project Folder Structure

```
ComicCraft_AI/
├── .env
├── .env.example
├── .gitignore
├── requirements-local-diffusion.txt
├── requirements.txt
├── ComicCraft - AI Comic Story Creator using Gemini Models.pdf
├── app/
│   ├── __init__.py
│   ├── config.py
│   ├── exporters.py
│   ├── gemini_flash.py
│   ├── gemini_pro.py
│   ├── image_generator.py
│   ├── layout_builder.py
│   ├── main.py
│   ├── routes.py
│   └── schemas.py
├── static/
│   ├── css/
│   │   └── style.css
│   └── exports/
│       └── comic_20260930_153431_523899.pdf
```

Installation and Configuration

- **Dependencies:** Python packages must be installed using `pip install -r requirements.txt` and `pip install -r requirements-local-diffusion.txt`.
- **Configuration:** The `.env.example` file must be copied to `.env` and populated with the necessary API keys and local configuration paths.

Module Implementation

Module 1: Application Core

Implemented via `app/main.py` and `app/routes.py`, establishing the web server and API/frontend endpoints. Data structures are validated using `app/schemas.py`. Configuration logic is housed in `app/config.py`.

Module 2: AI Generation

Implemented via `app/gemini_pro.py`, `app/gemini_flash.py`, and `app/image_generator.py`. The Gemini scripts handle prompt engineering and API calls to Google's models, while the image generator interfaces with the local diffusion setup.

Module 3: Assembly & Export

Implemented via `app/layout_builder.py` and `app/exporters.py`. The layout builder aligns the generated text and images, passing the structured data to the exporter, which writes the binary PDF file to the `static/exports/` directory.

Core Functionality

The core pipeline takes a user prompt, routes it to the Gemini models to generate a script and image prompts, passes the image prompts to the local diffusion model, and finally pipes all assets into the layout builder and exporter to render the PDF.

User Interface Implementation

The project features a web-based user interface, styled using CSS located at [static/css/style.css](#).

ComicCraft AI
Turn your idea into a five-panel comic.

Story Prompt
A brave fox exploring an enchanted forest

Main Character Name
Alex

Setting
forest

Story Tone
light-hearted

Art Style
anime

Generate Comic

Pipeline: Outline → Story & Dialogue → Illustrations → Layout → PDF

Figure: Web Interface

Integration

- **Backend and AI Services:** Direct API integration with Google Gemini Pro and Flash.
- **Backend and Image Generation:** Integration with a local diffusion model environment.

GitHub Repository: [REPO LINK](#)